

Appendix C. Important System Calls

These are some of the more important system calls to use when dealing with Linux. For most cases, however, it is best to use library functions rather than direct system calls, because the system calls were designed to be minimalistic while the library functions were designed to be easy to program with. For information about the Linux C library, see the manual at <http://www.gnu.org/software/libc/manual/>

Remember that `%eax` holds the system call numbers, and that the return values and error codes are also stored in `%eax`.

Table C-1. Important Linux System Calls

%eax	Name	%ebx	%ecx	%edx	Notes
1	exit	return value (int)			Exits the program
3	read	file descriptor	buffer start	buffer size (int)	Reads into the given buffer
4	write	file descriptor	buffer start	buffer size (int)	Writes the buffer to the file descriptor
5	open	null-terminated file name	option list	permission mode	Opens the given file. Returns the file descriptor or an error number.
6	close	file descriptor			Closes the give file descriptor

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%eax	Name	%ebx	%ecx	%edx	Notes
12	chdir	null-terminated directory name			Changes the current directory of your program.
19	lseek	file descriptor	offset	mode	Repositions where you are in the given file. The mode (called the "whence") should be 0 for absolute positioning, and 1 for relative positioning.
20	getpid				Returns the process ID of the current process.
39	mkdir	null-terminated directory name	permission mode		Creates the given directory. Assumes all directories leading up to it already exist.
40	rmdir	null-terminated directory name			Removes the given directory.
41	dup	file descriptor			Returns a new file descriptor that works just like the existing file descriptor.
42	pipe	pipe array			Creates two file descriptors, where writing on one produces data to read on the other and vice-versa. %ebx is a pointer to two words of storage to hold the file descriptors.

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%eax	Name	%ebx	%ecx	%edx	Notes
45	brk	new system break			Sets the system break (i.e. - the end of the data section). If the system break is 0, it simply returns the current system break.
54	ioctl	file de- scriptor	request	arguments	This is used to set parameters on device files. It's actual usage varies based on the type of file or device your descriptor references.

A more complete listing of system calls, along with additional information is available at <http://www.lxhp.in-berlin.de/lhpsyscal.html> You can also get more information about a system call by typing in `man 2 SYSCALLNAME` which will return you the information about the system call from section 2 of the UNIX manual. However, this refers to the usage of the system call from the C programming language, and may or may not be directly helpful.

For information on how system calls are implemented on Linux, see the Linux Kernel 2.4 Internals section on how system calls are implemented at http://www.faqs.org/docs/kernel_2_4/lki-2.html#ss2.11

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